Instructor Guide

Temporary Pacemaker Training Simulator — Scenario Reference & Debrief Companion

■ Training simulator only — not a medical device. All patient data is synthetic. This guide describes simulator behavior for instructional planning; it is not a clinical reference.

How the Simulator Works

C.A.R.E. SimPace pairs two devices over the local network. The **learner iPad** is the bedside — a vitals monitor with live ECG and pleth waveforms, alongside a full-function pacemaker generator (rate, output, sensitivity, AV delay, mode). The **instructor iPhone** is live control — load a scenario, set the underlying rhythm, inject faults, and watch the learner respond in real time. Both devices can also run solo, without pairing, for independent practice.

Every rhythm is driven by a physiology engine with clinically grounded capture thresholds, sensing behavior, and alarm conditions — not scripted animations. The numbers below are the exact values the simulator uses to judge learner performance.

Reference Thresholds

Parameter	Value	What it governs
Normal capture threshold	3.0 mA	V output needed to capture under normal conditions
Failure-to-capture threshold	12.5 mA	V output required while that fault is active
Battery-fault threshold	13.5 mA	V output required while battery depletion is active
Undersensed signal	1.2 mV	V/A sensitivity must be at or below this to sense the native beat
R-on-T window	0.22 – 0.40 s	Post-QRS interval where an unsensed spike induces V-Fib
RAP termination rate	≥ 320 ppm	Burst pace rate needed to convert atrial flutter to sinus
Runaway pacemaker rate	210 ppm	Rate the generator locks to when that fault is active

1 · Failure to Capture

Underlying rhythm: 3° AV Block, escape rate 33 bpm

Mode	LRL	A Output	V Output	V Sens	A Sens
DDD	80 ppm	5 mA	5 mA	2 mV	0.5 mV

THE CONCEPT

The generator is pacing, but ventricular output (5.0 mA) sits below the failure-to-capture threshold this fault imposes (12.5 mA) — pacing spikes appear on the ECG with no captured QRS following them. The patient remains dependent on their intrinsic 33 bpm escape rhythm.

WHAT THE LEARNER SHOULD DO

Recognize spikes without capture, then increase V Output using the V Out knob or the ADJ/step controls until the paced QRS reliably follows each spike.

HOW THE SIM DEFINES SUCCESS

The simulator logs "capture restored" once V Output reaches ≥ 12.5 mA — this is the same value used to gate the fault itself, so the fix is not arbitrary.

DEBRIEF QUESTIONS

- What did the ECG look like before the learner intervened? What does that pattern mean clinically (loss of capture vs. loss of output)?
- Why does a fresh post-op patient's threshold rise over the first 24–48 hours, and how does that map to what just happened here?
- What would you do at the bedside if increasing output didn't restore capture?

2 · Undersensing

Underlying rhythm: Normal Sinus Rhythm, 78 bpm

Mode	LRL	A Output	V Output	V Sens	A Sens
VVI	60 ppm	5 mA	6 mA	2 mV	0.5 mV

THE CONCEPT

V Sensitivity is set to 2.0 mV, but the undersensing fault drops the effective native-signal amplitude to 1.2 mV — below what the pacer can detect at that setting. The generator ignores the native rhythm and paces asynchronously into it, competing with the patient's own beats.

WHAT THE LEARNER SHOULD DO

Recognize the competition between paced and intrinsic beats, then lower V Sensitivity (more sensitive) until it is at or below 1.2 mV.

HOW THE SIM DEFINES SUCCESS

The simulator logs "sensing restored" once V Sensitivity reaches ≤ 1.2 mV, matching the undersensed-signal threshold the fault introduces.

DEBRIEF QUESTIONS

- What's the risk of leaving a pacer undersensing indefinitely? (This scenario's sequel — R-on-T — makes that risk concrete.)
- How would the ECG differ if this were oversensing instead?
- At what point should a learner escalate rather than keep adjusting settings?

3 · R-on-T → V-Fib Arrest

Underlying rhythm: Normal Sinus Rhythm, 75 bpm

Mode	LRL	A Output	V Output	V Sens	A Sens
VVI	71 ppm	5 mA	7 mA	2 mV	0.5 mV

THE CONCEPT

The highest-stakes scenario. Undersensing is active from the start, and the pacer's own clock and the native rhythm's clock are running independently. Roughly 8 seconds after Start, an unsensed pacing spike lands inside the 0.22–0.40 s post-QRS window — on the T wave — and the simulator induces true V-Fib.

WHAT THE LEARNER SHOULD DO

Recognize the undersensing (competing paced/native beats) and correct V Sensitivity to ≤ 1.2 mV **before** the ~8-second window closes.

HOW THE SIM DEFINES SUCCESS

If corrected in time, sensing restores and the vulnerable spike never fires on the T wave. If not, the monitor alarms “VF — DEFIBRILLATE,” BP and pleth flatline, and HR reads “--”.

DEBRIEF QUESTIONS

- Why is a spike landing on the T wave specifically dangerous, versus any other part of the cycle?
- How did it feel to work under a countdown? What does that simulate about real bedside urgency?
- If VF occurred, walk through the immediate next steps at the bedside — what happens after you defibrillate?

4 · Baseline Competency

Underlying rhythm: 3° AV Block, escape rate 33 bpm

Mode	LRL	A Output	V Output	V Sens	A Sens
DDD	80 ppm	6 mA	6 mA	2 mV	0.5 mV

THE CONCEPT

No fault is injected — this is the foundational “can you set up a pacer correctly” check. A post-op patient in complete heart block needs reliable dual-chamber pacing; the ECG will show the diagnostic AV-dissociation cue of independent P waves marching through the escape rhythm.

WHAT THE LEARNER SHOULD DO

Confirm DDD is selected, LRL/outputs are configured, and that both A-Pace and V-Pace are capturing reliably — essentially, verify a correct setup rather than fix a broken one.

HOW THE SIM DEFINES SUCCESS

There is no fault to “clear” here — success is a stable, correctly captured DDD-paced rhythm sustained through the scenario.

DEBRIEF QUESTIONS

- What does AV dissociation look like on this strip, and why is it the hallmark of 3° block?
- Why DDD instead of VVI for this patient?
- What would prompt you to reassess pacing settings even when everything looks fine?

5 · Atrial Flutter — RAP

Underlying rhythm: Atrial Flutter, 150 bpm

Mode	LRL	A Output	V Output	V Sens	A Sens
DDD	60 ppm	6 mA	6 mA	2 mV	0.5 mV

THE CONCEPT

A re-entrant atrial flutter circuit is running at 150 bpm. Rapid Atrial Pacing (RAP) can interrupt the re-entrant loop if the burst rate is fast enough — too slow, and the circuit simply re-establishes itself.

WHAT THE LEARNER SHOULD DO

Open the RAP tab on the pacemaker generator, set a burst rate of ≥ 320 ppm, and hold to deliver the burst.

HOW THE SIM DEFINES SUCCESS

The simulator converts the rhythm to sinus only when the delivered RAP rate meets or exceeds 320 ppm — lower rates will not terminate the flutter.

DEBRIEF QUESTIONS

- Why does a faster burst rate succeed where a slower one fails, mechanistically?
- What are the risks of rapid atrial pacing, and why is it typically reserved for monitored settings?
- How would you counsel a patient about recurrence risk after successful termination?

Fault Injection Reference

Beyond the five built-in scenarios, instructors can inject any of the following faults on top of any rhythm from the Instructor console, for improvised or advanced sessions:

Fault	What it does
Failure to Capture	Raises the V output threshold needed to capture (see Reference Thresholds)
Undersensing	Pacer ignores native beats below the undersensed-signal threshold
Oversensing (EMI)	Pacer inappropriately inhibits, mistaking noise for native activity
Lead Fracture	Intermittent capture — roughly half of paced beats fail at random
Battery Depletion	Output threshold rises further and a low-battery indicator appears
Loss of Output	Generator stops pacing entirely despite being powered on
Runaway Pacemaker	Generator locks to a rapid, uncontrolled 210 ppm pacing rate
PMT	Pacemaker-mediated tachycardia — paces up toward the programmed upper rate